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Professional Profiles

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Objectives

Goal-oriented BSCS Graduate with a 4.0/4.0 academic record, offering a deep technical stack in **Data Science, Machine Learning, and Cloud Computing**. I specialize in extracting value from complex datasets and building purposeful, AI-driven solutions for enterprise environments. With a background in Web and Mobile development, I bring a holistic approach to IT—combining rigorous data analysis with hands-on software engineering to solve high-stakes organizational problems.

Education

Bachelor of Science (BS) in Computer Science : COMSATS University Islamabad, Sahiwal Campus | 2022 – 2026

Academic Performance: CGPA: 4.0/4.0 (**Patron's Gold Medal**)

Selected Coursework: Object Oriented Programming, Information Security, Artificial Intelligence, Data Science, Compiler Construction, Machine Learning, Software Engineering Concepts, Operating Systems, Data Structures & Algorithms, Parallel and Distributed Computing

Intermediate: Punjab College Okara

Academic Performance: 93.8% (1032/1100)

Matriculation: Govt. Higher Secondary School Renala Khurd

Academic Performance: 92.6% (1019/1100)

Languages

Urdu: Native proficiency

English: Intermediate - B1

Technical Skills

Programming Languages: Python, R, Kotlin, NodeJS, C++, C, SQL, Bash scripting, Assembly Language

AI/ML & Data: Scikit-learn for machine learning, NumPy, Data processing with Pandas, Data visualization with Matplotlib

Frameworks & Tools: Android Studio, Firebase, .NET development (Windows Forms), Parallel programming with OpenMP, Git repository management, NetBeans, Visual Studio, Power BI, Looker Studio, Google Colab

Web: HTML5, CSS3, JavaScript, PHP, React JS, Xampp Server

Academic Projects

Final Year Project (FYP) Corinna AI – AI-driven Sales Agent & SaaS Platform

Developed an AI-powered SaaS marketing platform with an automated sales representative for real-time customer engagement and lead generation.

Integrated an intelligent chatbot for FAQ handling, automated appointment scheduling, and secure payment processing

Tools & Technologies: Next.js, TypeScript, OpenAI API, Prisma, PostgreSQL, Clerk, Stripe, Pusher, Tailwind CSS.

Automated Web Scraping Suite – Data Extraction Tools: Developed specialized scripts to bypass Cloudflare protections and scrape Telegram links from Coingecko and Dex Screener. Engineered automation tools for extracting and organizing academic contact data from international university websites.

Tools & Technologies: Python, BeautifulSoup, Cloudscraper, Tor Browser.

Plant Disease Classification System

Developed a CNN-based classifier to identify diseases in plant leaves using Python and TensorFlow.

Designed full preprocessing pipelines and image augmentation to enhance model training and evaluation.

Tools & Technologies: Python, TensorFlow, OpenCV, Google Colab, VS Code.

Emotion Detection System

Engineered a multi-modal system to classify human emotions from text, audio, and facial expressions.

Implemented and benchmarked Supervised, Unsupervised, and Reinforcement Learning models including CNNs and RNNs.

Tools & Technologies: Python, scikit-learn, TensorFlow, OpenCV, NLTK, librosa

Federated Learning System (Parallel & Distributed Computing)

Created a prototype for decentralized model training to demonstrate secure, privacy-preserving machine learning.

Built a client-server architecture using multi-threading and message passing in a distributed environment.

Tools & Technologies: Python, PyTorch, multiprocessing, socket programming, NumPy.

Power BI Dashboard – Social Media Insights

Constructed a dynamic dashboard to visualize key KPIs like engagement rates, impressions, and audience demographics.

Integrated diverse data sources including CSV, Excel, and APIs to create interactive visual reports.

Hospital Management System – Record Management Platform: Developed a web-based system designed to efficiently maintain and manage patient medical records.

Tools & Technologies: C++, HTML, CSS.

Gym Management System – Fitness Tracking Utility: Created a management tool to streamline the tracking and maintenance of trainee registration and records.

Tools & Technologies: C++.

AgriSol – Agricultural Services Website: Full-stack web platform connecting farmers with agritech services; features user auth, service listings, query forms and admin dashboard.
Tools & Technologies: HTML, CSS, JavaScript, PHP/Node.js, MySQL/Firebase

Digital Ecosystem & Service Applications

Developed a suite of mobile applications using **Kotlin**, **Room/SQLite**, and **Firebase** for real-time data management.

Donation App: Engineered a platform to facilitate secure and efficient financial donations between donors and receivers.

Society Services App: Created a community-focused application connecting users with local service providers to streamline service requests.

Tools & Technologies: Kotlin, Android Studio, Firebase, Room Database, SQLite.

Windows Forms LL(1) & SLR Parser (Compiler Construction Project)

Implemented grammar analysis tools and parsing simulators (LL(1), SLR) using C# and Windows Forms.

Tools & Technologies: C#, Visual Studio

Cloud Storage & File Sharing System

Implemented a cloud-based storage system to upload, store, and retrieve files securely. Demonstrated data availability, scalability, and access control in a cloud environment.

Tools & Technologies: Google Drive / AWS S3 (conceptual), REST APIs, HTML, CSS, JavaScript.

Supervised Learning – Student Performance Prediction

Developed a machine learning model to predict student academic performance based on attendance, study hours, and previous results.

Applied data preprocessing, feature selection, and model evaluation techniques.

Tools & Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib.

Natural Language Processing – Sentiment Analysis

Implemented an NLP-based sentiment analysis system to classify text as positive, negative, or neutral.

Applied text preprocessing, vectorization, and machine learning classifiers.

Tools & Technologies: Python, NLTK, Scikit-learn, TF-IDF, Pandas.

Certifications

Python Developer: Coursera (University of Michigan, USA) | Sep 2023 – Dec 2023

Event Management: Computer Science Society | July 2023 – July 2024

Oracle: Oci Generative Certification (Exam Applied)

Work Experience

Machine Learning & Cloud Computing

TECVEQ Remote

Duration: 6 Months

Worked as a Machine Learning and Cloud Computing-based Developer, contributing to multiple projects involving data-driven solutions and cloud-based architectures. Responsibilities included developing and deploying machine learning models, assisting in cloud infrastructure setup, and optimizing application performance in a distributed environment. Collaborated remotely with the team, followed project requirements, and delivered tasks within flexible schedules and deadlines.

Achievements Activities

Academic Excellence: Consistently maintained a perfect 4.0 CGPA throughout BS Computer Science.

Technical Breadth: Successfully completed 20+ micro-projects ranging from system utilities (PDF Merging, OS Modules) to interactive games.

Interests: Machine Learning, and Data Science.

Hobbies: Photography and Gaming.

References

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